

PAWAN PATIL

AI Product Engineer | Forward Deployed Engineer | Frontend & GenAI Systems

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GitHub: github.com/Pawanp-23 | Portfolio: portfolio-pawan-ten.vercel.app

PROFESSIONAL SUMMARY

B.Tech IT student with an Electronics & Telecommunication foundation, building AI-enabled products, Retrieval-Augmented Generation (RAG) systems, real-time voice agents, and production-ready React applications. Hands-on in full-stack development, REST APIs, LLM APIs, vector search, GIS, and cloud deployment, with a record of leading teams and shipping live products. Targeting AI Product Engineering, Forward Deployed Engineering, and GenAI roles.

TECHNICAL SKILLS

AI / GenAI: LLM APIs, RAG, Embeddings, Vector Search, Prompt Engineering, AI Agents, Voice AI, ML, SHAP
Frontend: React.js, TypeScript, JavaScript, Vite, Tailwind CSS, Framer Motion, Responsive UI, Component Architecture
Backend / Data: Python, FastAPI, Node.js, Express.js, REST APIs, MongoDB, Atlas Vector Search, ODMantic, Motor
Real-time / Voice: Deepgram STT, Cartesia TTS, WebRTC, WebSockets, Silero VAD, Gemini Live API, AudioWorklet
Tools / Deployment: Git, GitHub, Vercel, Render, API Integration, System Integration, Debugging, JSON, GeoJSON
Systems / Other: QGIS, OpenStreetMap, MATLAB, C, C++, Embedded C++, Arduino Uno, 8051 Microcontroller

EXPERIENCE

Project Lead / App Lead — Maharashtra Remote Sensing Application Centre (MRSAC), Nagpur **Jun 2026 – Jul 2026**

- Led React.js development and software architecture for a GIS-based EV Smart Platform; owned the alerts, user profile, station discovery, service-centre lookup, and analytics modules.
- Integrated field-verified GIS data covering **8 charging stations, 34 charging points, and 5 battery service centres**; delivered reusable GeoJSON/Shapefile datasets and a functional web platform.

Industrial Intern — Bharat Sanchar Nigam Limited (BSNL)

May 2025 – Jun 2025

- Rotated across NIB, FTTH, Routing & Networking, and Sales & Marketing; gained practical exposure to optical-fiber communication, broadband infrastructure, network architecture, and service operations.

Diploma Engineer Trainee — Subros Limited, Pune

Jan 2024 – Jun 2024

- Worked across Quality Control, Water Leak Testing, HVAC Assembly, and Production; performed quality inspections, component validation, and manufacturing checks under 5S and safety standards.

PROJECTS

Clinic Voice AI + RAG | AI Voice Agent, Knowledge Retrieval

Python, OpenAI API, WebRTC, MongoDB Atlas Vector Search | Live: citycare-frontend-wlw8.onrender.com

- Designed a real-time voice AI pipeline using Deepgram speech-to-text, Cartesia text-to-speech, OpenAI LLM processing, WebRTC streaming, and Silero voice activity detection.
- Implemented RAG with embeddings and MongoDB Atlas Vector Search to deliver grounded, knowledge-based conversational responses for a clinic assistant.

Whitfield Warehouse Management System | Full-Stack Web Application

React.js, REST APIs, Render | Frontend: whitfield-frontend.onrender.com | API: whitfield-api.onrender.com

- Built and deployed a full-stack warehouse management system for inventory visibility, warehouse operations, and stock-movement workflows; integrated the React.js frontend with REST API services.

AI-Based Predictive Personnel Stress & Welfare Monitoring System

Smart India Hackathon

Team Lead | Machine Learning, SHAP, RAG, REST APIs, React.js

- Leading system architecture across ML risk scoring, SHAP explainability, feature engineering, and RAG/API integration; driving privacy-aware welfare recommendations, frontend delivery, research, and technical documentation.

Other Live Builds: LRN Readiness (lrn-client.vercel.app) | Team Codex Canteen (team-codex-canteen.vercel.app)

LEADERSHIP & PRODUCT COLLABORATION

AI Product / Frontend Contributor — EIGI AI | fde.eigi.ai

2026

- Translate ambiguous product requirements into structured UI flows and polished, API-integrated React.js interfaces; ship iterative FDE-oriented releases with backend and leadership stakeholders.

Coordinator — Advantage Vidarbha Startup Summit

Nagpur

- Coordinated event operations and stakeholder interactions with founders, investors, entrepreneurs, and industry leaders.

EDUCATION

B.Tech, Information Technology — St. Vincent Pallotti College of Engg. & Technology, Nagpur

SGPA 7.0 | In Progress

Diploma, Electronics & Telecommunication Engg. — Government Polytechnic, Nagpur | 82%

2024

Class X — Somalwar High School, Nagpur | 78%

2021

CERTIFICATIONS

MATLAB Onramp (MathWorks) | TCS iON Young Professional | MS-CIT | College Hackathon Certification